UCI Credit Card Default Dataset: Financial Interpretation Guide

(Explanation of PAY_x, BILL_AMTx, PAY_AMTx, and Default Y)

1. Overview

The UCI Credit Card Default dataset contains financial behavior of 30,000 credit card clients over a 6-month period. The dataset includes information about credit limits, repayment behavior, bill amounts, payment amounts, and whether the client defaulted the following month.

This document summarizes the correct financial interpretation of all variables that students often find confusing.

2. Key Variables

- **PAY_x** — Repayment status (months overdue)
- **BILL_AMTx** — Monthly bill amount (amount due)
- **PAY_AMTx** — Actual payment amount
- **Y (default payment next month)** — Whether the client defaulted next month

Each of these represents a different part of the credit card billing and repayment cycle.

3. Meaning of PAY_x (Repayment Status Codes)

The variables PAY_0, PAY_2, PAY_3, PAY_4, PAY_5, PAY_6 represent the client's **repayment status** in the most recent 6 months. PAY_0 is the current month's status; PAY_2 corresponds to one month earlier; PAY_6 corresponds to five months earlier.

PAY_x Value Interpretation

Value	Meaning
-2	No consumption / No bill issued
-1	Paid duly (no delay)
0	Paid on schedule
1	1 month overdue
2	2 months overdue
3	3 months overdue
⋮	⋮
8	8 months overdue (maximum category)

Important Notes

- PAY_x is a **monthly snapshot**, not a cumulative count.
- Overdue months do **not** automatically decrease by 1 each month.
- Even if PAY_0 = 3, PAY_2 does not have to be 2.
- The overdue month count can decrease only if the client pays **at least the required minimum payment**.
- Partial payments do **not** reduce the overdue months. For example, paying "half" of a monthly amount does not change the overdue count.
- When a payment is made, it is always applied to the **oldest** unpaid balance first (FIFO principle).

4. Why PAY_1 Does Not Exist

Originally, financial systems had both PAY_0 and PAY_1. But since both referred to the most recent billing cycle, the dataset provider removed PAY_1 as it was redundant. Thus the variables jump from PAY_0 directly to PAY_2.

5. Meaning of BILL_AMTx (Monthly Bill Amount)

BILL_AMT1 through BILL_AMT6 represent the **actual amount billed by the credit card company** in each month.

- BILL_AMT1 — this month's bill (most recent)
- BILL_AMT2 — last month's bill
- BILL_AMT3 — two months earlier
- ...
- BILL_AMT6 — five months earlier

What is included in a bill amount?

- Previous unpaid balance (rollover)
- New purchases
- Interest charges
- Penalties (if any)

Thus, BILL_AMT values reflect the real financial behavior of the client, including spending and unpaid balances.

6. Meaning of PAY_AMTx (Amount Paid)

PAY_AMT1 through PAY_AMT6 represent the **actual amount of money the client paid** to the card company in each month.

- PAY_AMT1 — amount paid this month
- PAY_AMT2 — amount paid last month
- PAY_AMT3 — amount paid two months earlier
- ...
- PAY_AMT6 — amount paid five months earlier

Important Notes

- PAY_AMT is an **absolute payment amount**.
- It can be zero if the client did not pay anything that month.
- A partial payment (e.g., 10% or 50% of the balance) does **not** reduce overdue months unless it meets or exceeds the minimum payment required.

7. FIFO Principle in Overdue Repayment

When the client makes a payment:

- Payments are always applied to the **oldest** unpaid balance first.
- Overdue months decrease only when the oldest overdue amount has been fully paid.

Example:

- If a client owes 100 (3 months ago), 10 (2 months ago), and 50 (1 month ago) and pays 10 this month:
- The 10 is applied to the 3-month-old balance.
- If the 3-month-old balance is not fully paid off, the overdue month count does not decrease.

8. Meaning of Y (Default Payment Next Month)

The variable *default payment next month* indicates whether the client defaulted in the next billing cycle.

Value	Meaning
0	No default (paid the minimum amount by due date)
1	Default (failed to pay the minimum amount)

Key Points

- Default does not mean the client rolled over a balance.
- Default means the client did **not pay the minimum required amount** on time.
- PAY_x values often strongly correlate with Y.

9. Summary

- PAY_x represents monthly overdue status (integer code).
- BILL_AMTx represents actual bill amounts.
- PAY_AMTx represents the actual amount paid.
- Overdue months only decrease when full minimum payment is made.
- Payments are applied to the oldest unpaid balance (FIFO).
- Y indicates next month's default (minimum payment not made).